

Gyro-Enhanced Micromouse

Project report — embedded sensing and closed-loop control

Overview

A compact autonomous vehicle combining inertial sensing and time-of-flight ranging to follow a maze while correcting accumulated heading error.

My Role

Designed the sensing and control approach, characterised gyro bias, and integrated the feedback loop with navigation logic.

Technical Approach

- ICM-20602 gyroscope for angular-rate estimation
- VL53L1X time-of-flight sensors for wall distance
- Bias compensation and closed-loop heading control
- Simulink modelling before hardware testing

Outcome

A more stable trajectory with reduced drift and more repeatable maze navigation.